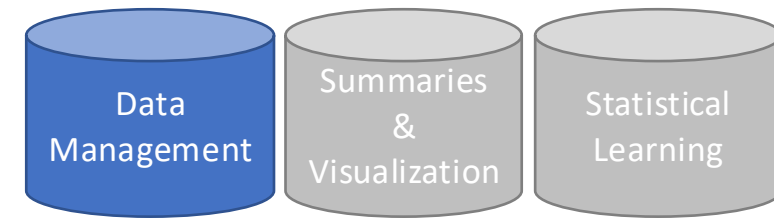


# Technology Agnostic

Teach concepts before specific technology

# Technology Agnostic Data Verbs



| D | G            |
|---|--------------|
| 5 | <del>a</del> |
| 3 | <del>a</del> |
| 4 | <del>a</del> |
| 2 | <del>a</del> |
| 3 | <del>a</del> |
| 5 | <del>a</del> |
| 4 | <del>a</del> |

**Select (D)**

| D |
|---|
| 5 |
| 3 |
| 4 |
| 2 |
| 3 |
| 5 |
| 4 |

| D            | G            |
|--------------|--------------|
| <del>5</del> | <del>a</del> |
| <del>3</del> | <del>a</del> |
| 4            | b            |
| 2            | b            |
| 3            | c            |
| 5            | c            |
| 4            | c            |

**Filter**  
(G is not a)

| D | G |
|---|---|
| 4 | b |
| 2 | b |
| 3 | c |
| 5 | c |
| 4 | c |

| D | G |
|---|---|
| 5 | a |
| 3 | a |
| 4 | b |
| 2 | b |
| 3 | c |
| 5 | c |
| 4 | c |

**Mutate**  
( $N = 2 * D$ )

| D | G | N  |
|---|---|----|
| 5 | a | 10 |
| 3 | a | 6  |
| 4 | b | 8  |
| 2 | b | 4  |
| 3 | c | 6  |
| 5 | c | 10 |
| 4 | c | 8  |

| D | G |
|---|---|
| 5 | a |
| 3 | a |
| 4 | b |
| 2 | b |
| 3 | c |
| 5 | c |
| 4 | c |

**Sort (D, G)**

| D | G |
|---|---|
| 2 | b |
| 3 | a |
| 3 | c |
| 4 | b |
| 4 | c |
| 5 | a |
| 5 | c |

# Exercise 1

## Purchasing Snowshoes

### Data Verbs: Activity #1

[ [Paper Version](#) ]

Consider the following sequence of structured data below. Identify the data verb for the transition between each structured data table.

**Goal:** Obtain a list of locations that sell snowshoes on a Sunday near Winona, MN. The list should be ordered by distance and near Winona is defined to be locations whose area code is 507 or 608.

**Notes:** 1) Your browser is a canvas for this task. 2) With your mouse, drag and drop the appropriate data verb for each table transition. 3) Items can be resized with mouse + Shift. 4) Page will autoscroll when dragging data verbs.

Table 0

| Name                              | Address                                      | Phone        | Distance (miles) | Open Sundays       | Rating | Notes       |
|-----------------------------------|----------------------------------------------|--------------|------------------|--------------------|--------|-------------|
| Wild River Ski Rental & Sales     | 39797 Park Trail, Center City, MN 55012      | 651-257-0685 | 115.68           | Yes                |        |             |
| Fitzharris Ski, Bike, and Outdoor | 105 7th Ave S, St Cloud, MN 56301            | 320-251-2844 | 161.88           | No                 | 4.7    |             |
| Welch Village                     | 26685 County 7 Blvd, Welch, MN 55089         | 651-258-4567 | 64.55            | Yes (Dec-Mar only) | 4.5    | Rental Only |
| Dunham's Sports                   | 140 Tyler Rd N, Red Wing, MN 55066           | 651-388-1402 | 59.12            | Yes                | 4.1    |             |
| Scheels                           | 1850 Adams St Suite 6, Mankato, MN 56001     | 507-386-7767 | 115.22           | Yes                | 4.5    |             |
| Cabela's                          | 3900 Cabela Dr NW, Owatonna, MN 55060        | 507-451-4545 | 80.14            | Yes                | 4.4    |             |
| Tyrol Ski & Sports                | 1923 2nd St SW, Rochester, MN 55902          | 507-288-1683 | 42.34            | No                 | 4.6    |             |
| Scheels                           | 1220 12th St SW, Rochester, MN 55902         | 507-281-2444 | 41.75            | Yes                | 4.6    |             |
| Myre-Big Island State Park        | 19499 780th Ave, Albert Lea, MN 56007        | 507-668-7060 | 87.85            | Yes                | 4.5    | Rental Only |
| Dunham's Sports                   | 2500 Bridge Ave, Albert Lea, MN 56007        | 507-369-1444 | 89.17            | Yes                | 4.1    |             |
| Whitewater State Park             | 19041 MN-74, Altura, MN 55910                | 507-312-2300 | 20.17            | Yes                | 4.8    | Rental Only |
| Brone's Bike Shop                 | 615 S Main St #35, Fountain City, WI 54629   | 608-687-8601 | 6.26             | No                 | 4.9    |             |
| Adventure Cycle & Ski             | 178 Center St, Winona, MN 55987              | 507-452-4228 | 0.43             | No                 | 4.9    |             |
| Outdoor Connection                | 1601 Farwell St, La Crosse, WI 54601         | 608-785-8860 | 25.88            | Yes                | 4.8    | Rental Only |
| Island Outdoors                   | 1330 Bainbridge St, La Crosse, WI 54603      | 608-782-5562 | 23.64            | Yes                | 4.8    |             |
| Redfeather Snowshoes              | 2700 Commerce St, La Crosse, WI 54603        | 800-525-0081 | 24.37            | No                 | 5      |             |
| Dick's Sporting Goods             | 4400 WI-16, La Crosse, WI 54650              | 608-406-3684 | 25.46            | Yes                | 3.7    |             |
| Decorah Bicycles                  | 101 College Dr, Decorah, IA 52101            | 563-382-8209 | 51.88            | Yes                | 5      |             |
| CrawDaddy Outdoors                | 207 E Bremer Ave, Waverly, IA 50677          | 319-352-9129 | 100.33           | No                 | 4.7    |             |
| Hoigaards                         | 5425 Excelsior Blvd, St Louis Park, MN 55416 | 952-929-1351 | 104.01           | Yes                | 3.9    |             |
| Midwest Mountaineering            | 309 Cedar Ave, Minneapolis, MN 55454         | 612-339-3433 | 101.74           | Yes                | 4.7    |             |
| REI                               | 1955 County Rd B2 W Schneiderman's Plaza,    | 651-635-0211 | 101.35           | Yes                | 4.5    |             |
| ERIK'S Bike Board Ski             | 1278 Town Centre Dr Ste 170, Eagan, MN 55121 | 651-209-6046 | 92.54            | Yes                | 4.1    |             |
| Gear West Ski and Bike            | 1786 Wayzata Blvd, Long Lake, MN 55356       | 952-473-0377 | 115.08           | Yes                | 4.6    |             |
| Black Apple Sports                | 1255 Town Centre Dr, Eagan, MN 55123         | 651-230-0002 | 92.46            | Yes                | 4.2    |             |

**FILTER(**  
Area Code =  
{507, 608} )

**FILTER(**  
No Rental)

**FILTER(**  
Open Sundays)

**MUTATE(**  
LEFT,3)

**SELECT(**  
Name,Address,  
Phone)

**SELECT(**  
Name,Address,  
Phone,Distance,Rating  
)

<http://statsclass.org/dsci210/Dataverbs/>

# Low Code

Start with technology that minimizes the  
need for code

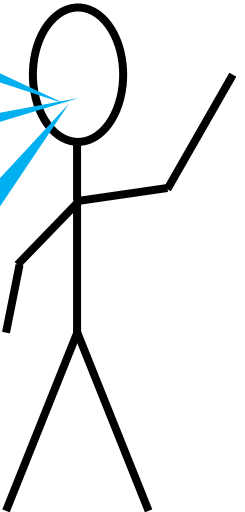
# Learning to program is hard!

```
(survey
  %>% pivot_longer(starts_with('F'),
    names_to = "question",
    values_to = "response")
  %>% inner_join(coding, by = join_by(question == Column.Name))
  %>% mutate(reg_coding = case_match(response,
    "Strongly Disagree" ~ 1,
    "Somewhat Disagree" ~ 2,
    "Neither Agree nor Disagree" ~ 3,
    "Somewhat Agree" ~ 4,
    "Strongly Agree" ~ 5
  ),
    rev_coding = 6 - reg_coding,
    coded_response = ifelse(Needs.Reverse.Coding. == 'Yes',
      rev_coding,
      reg_coding),
    type = str_split_i(question, '\\.', 1)
  )
  %>% select(ID, type, coded_response)
  %>% pivot_wider(id_cols = ID,
    values_from = coded_response,
    names_from = type,
    values_fill = 0,
    values_fn = sum)
)
```

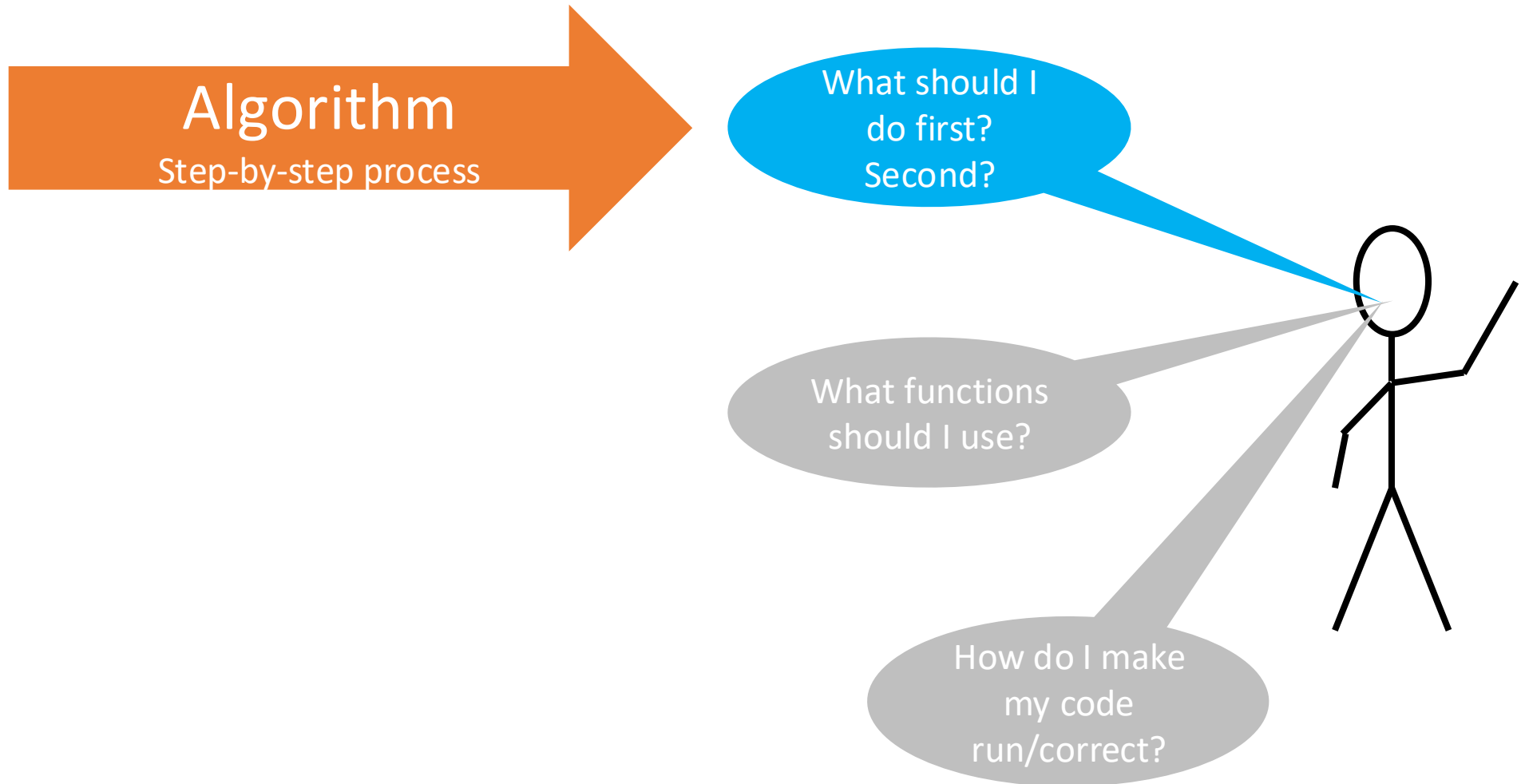
What should I  
do first?  
Second?

What functions  
should I use?

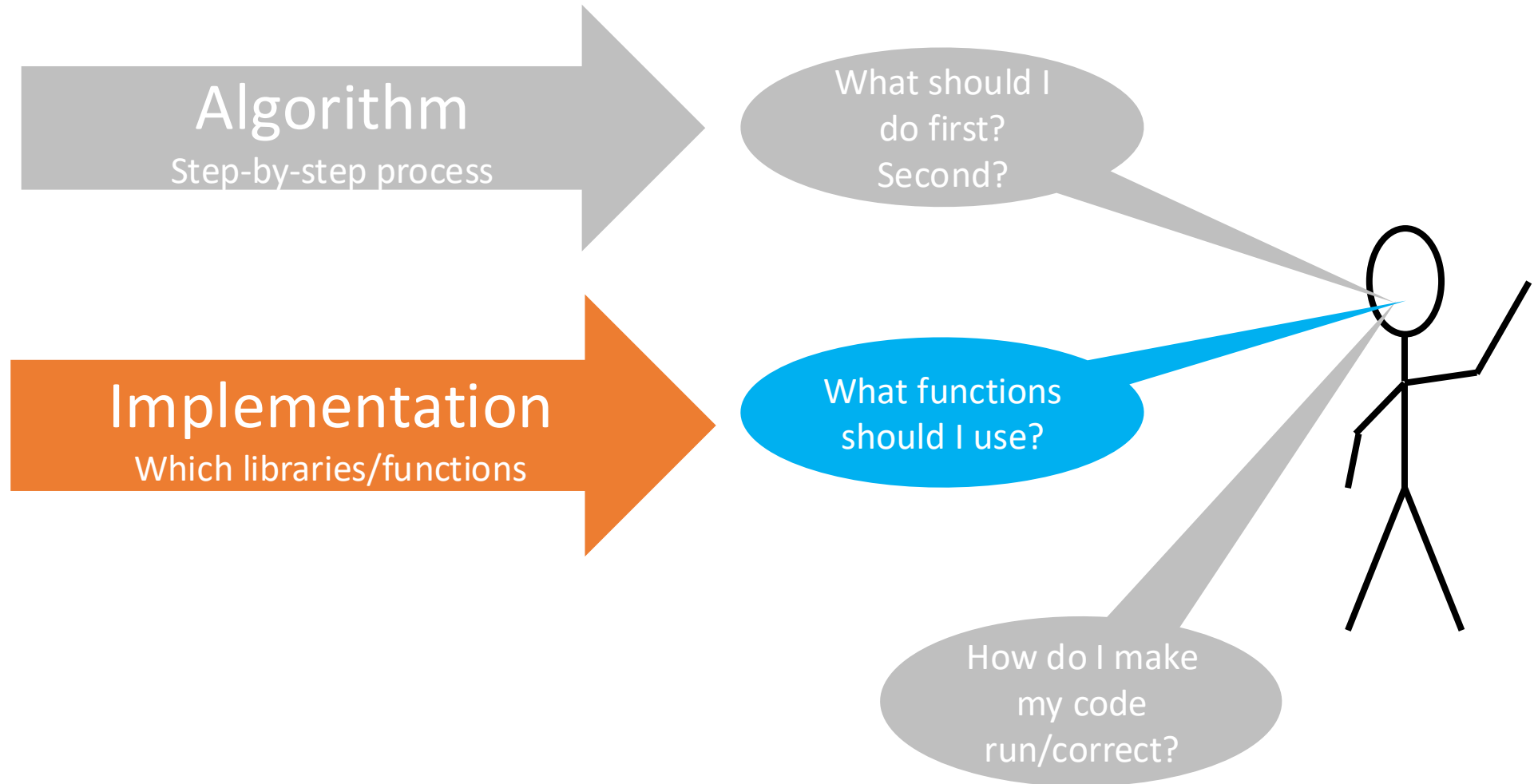
How do I make  
my code  
run/correct?



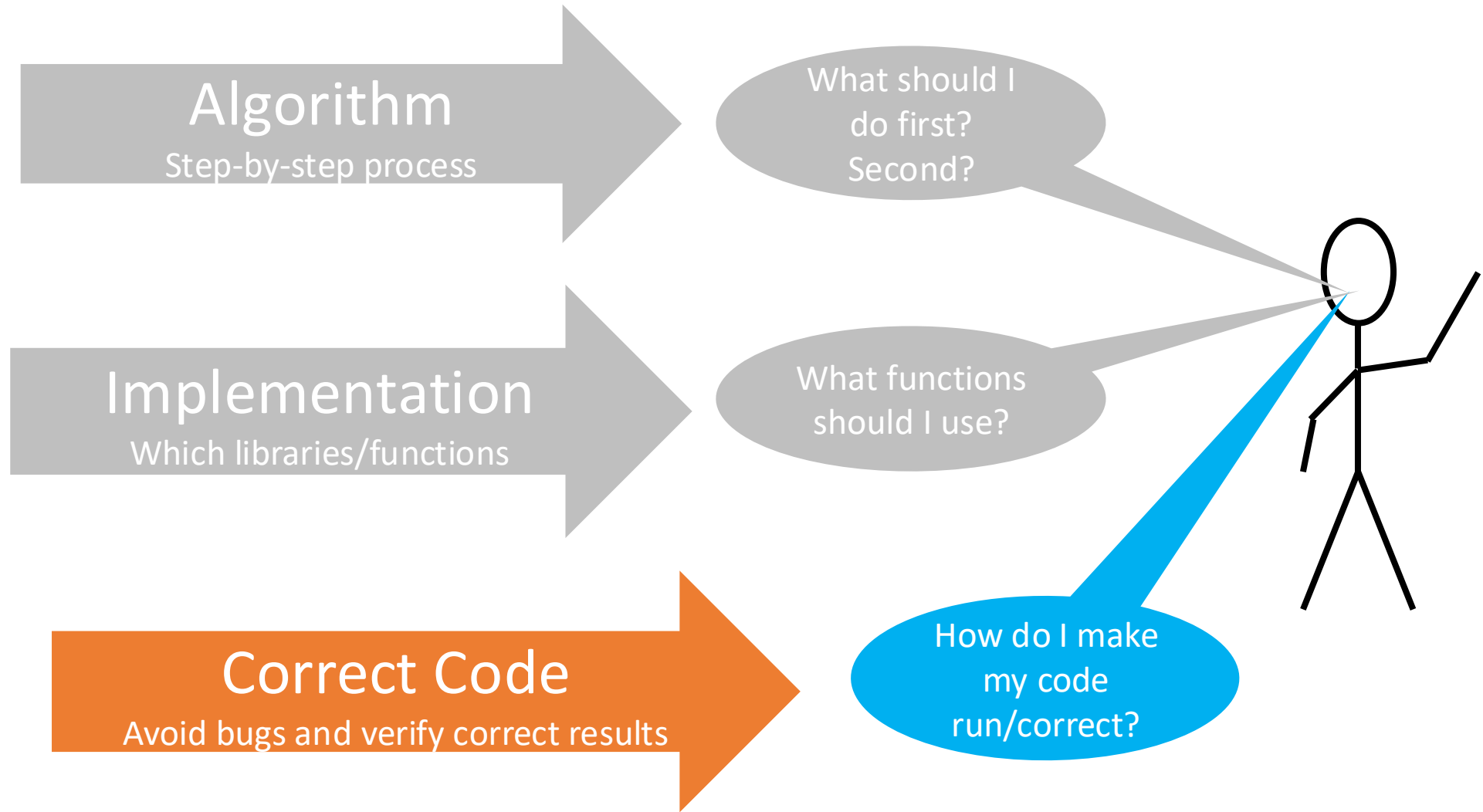
# Doug is struggling!



# Doug is struggling!



# Doug is struggling!





We decouple the algorithm and the code!

# The algorithm is most important

Let's make sure Doug is proficient here first!

Start  
here!



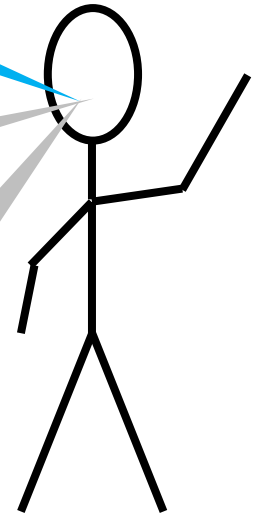
What should I  
do first?  
Second?



What functions  
should I use?



How do I make  
my code  
run/correct?



# Coding should come later in the first course

## Outline for DSCI 210

1. Intro to Tidy Data
2. Aggregation
3. Data Verbs
4. Intro to Visualization
5. Classification and CART
6. String Verbs and Window Functions

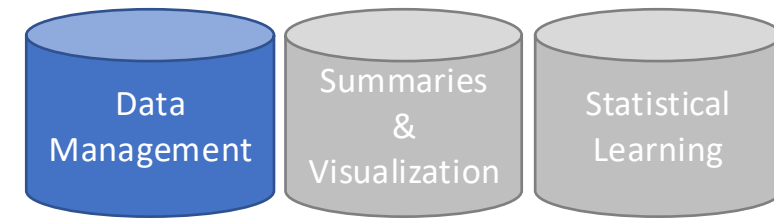
Low Code

7. Managing data with R/tidyverse
8. Reshaping tables
9. Table joins
10. Intro to Arules

Scripting + Low Code

# Low Code

## Data management with Tableau Prep



```
IF [Needs Reverse Coding?] == "Yes"
THEN [Reverse code]
ELSE [Regular code]
END
```

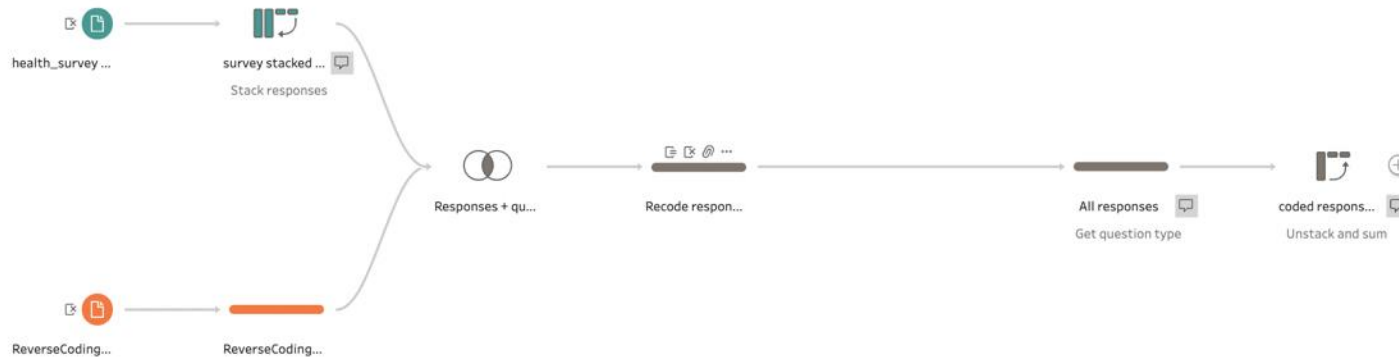
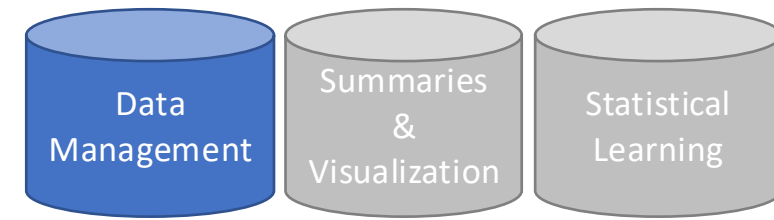
Only Code

# Translate to Code

Implementing code from low code  
solutions

# Translate to Code

## From Tableau Prep to dplyr



```
rvey
>% pivot_longer(starts_with('F'),
  names_to = "question",
  values_to = "response")
%% inner_join(coding, by = join_by(question == Column.Name))
%% mutate(reg_coding = case_match(response,
  "Strongly Disagree" ~ 1,
  "Somewhat Disagree" ~ 2,
  "Neither Agree nor Disagree" ~ 3,
  "Somewhat Agree" ~ 4,
  "Strongly Agree" ~ 5
),
  rev_coding = 6 - reg_coding,
  coded_response = ifelse(Needs.Reverse.Coding. == 'Yes',
    rev_coding,
    reg_coding),
  type = str_split_i(question, '\\.', 1)
)
%% select(ID, type, coded_response)
%% pivot_wider(id_cols = ID,
  values_from = coded_response,
  names_from = type,
  values_fill = 0,
  values_fn = sum)
)
```